

Route 3: the non-lattice remainder is non-load-bearing for B5's admissibility-bounded statement (Lemma 2 covers it)

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

After the T7 re-proof attempt (R-036) closed only the poly-separated subclass, route 3 of its next-action list asks: is H-NONLATTICE-REMAINDER-EXCLUDED actually load-bearing for B5's OWN statement, or – as for the published C_{full} head (synthesis Sec. 4b) – is the non-lattice remainder already covered by the elementary coherence circle-packing Lemma 2? **Scope:** the dependence of B5's beyond-layer bound on the competitor's lattice structure. **Not claimed:** any tier/gate/hypothesis flip (operator decides); any new theorem (Lemma 2 already exists, res5_036).

2. B5's beyond-layer bound depends on the competitor only through (T', n)

The admissible (coherence-resolved) competitor class fixes: real antipodal $Q = -Q$; packing-bounded $|Q| = n \leq n_{\text{pack}} = 16/\theta_{\min}^2$; coherence- resolved, pairwise geodesic $\geq \theta_{\min}$. The beyond-layer machinery enters the comparison through three competitor-dependent quantities, EACH a function of $(T'(Q), n)$ only:

$$K_{\text{floor}} \leq T'(Q) \text{ (Lemma A),} \quad R_{\text{lead}} = 23.2(1 + T')I,$$

$$K(n) \leq 8 + 4\sqrt{14}\kappa^4\sqrt{n} \text{ (rectangle, pattern-generic),}$$

where $T'(Q) = \max_{t \neq 0} \#(Q \cap C_t)$ is the maximal proper sum-circle occupancy. No other competitor feature enters; in particular the lattice/non- lattice distinction enters ONLY via T' .

3. Lemma 2 caps T' pattern-genericity (lattice and non-lattice alike)

Lemma 1 (Coherence circle-packing, res5_036; restated). *For an admissible Q and any $t \neq 0$, $\#(Q \cap C_t) \leq \lfloor 2\pi/\theta_{\min} \rfloor$.*

Proof (pure geometry). C_t is a Euclidean circle of radius $\rho = \sqrt{q_0^2 - |t|^2/4} \leq q_0$. Its Q -points are pairwise geodesically $\geq \theta_{\min}$, hence pairwise Euclidean $\geq d = 2q_0 \sin(\theta_{\min}/2)$. The k cyclically-ordered points have a smallest adjacent gap $\Delta_{\min} \leq 2\pi/k$; coherence forces $d \leq 2\rho \sin(\Delta_{\min}/2) \leq 2\rho \sin(\pi/k)$, so with $\rho \leq q_0$, $k \leq \pi/\arcsin(\sin(\theta_{\min}/2)) = 2\pi/\theta_{\min}$. $\square$

Key point: the proof uses ONLY $(\theta_{\min}, q_0)$ – no lattice, arithmetic, divisor, or decoupling input. Across $\mathcal{R}$, $\theta_{\min} \geq 0.596$, so the cap is ≤ 10 (machine: assert 1, sharp value 10 on the largest circle $\rho = q_0$; assert 2, monotone in ρ and never above the cap; assert 3, the idealized equatorial packing meets the cap while the realized full-sphere admissible max is $8 < 10$ per the head's adversarial search; assert 4, random NON-LATTICE $\theta_{\min}$ -separated antipodal configs obey the cap). The bound is identical for lattice and non-lattice competitors.

4. Consequence: the non-lattice remainder is non-load-bearing

Combining §2 and §3: within the admissible class, $T' \leq 10$ (Lemma 2) and $n \leq n_{\text{pack}} \approx 49$ are BOTH bounded pattern-generically, so

$$R_{\text{lead}} = 23.2 \cdot 11 \cdot I \leq 0.510 < 1, \quad K(n_{\text{pack}}) \leq 8 + 4\sqrt{14}\sqrt{49} \approx 113 \ll 5972 = K_{\text{budget}}$$

hold for EVERY admissible competitor, lattice OR non-lattice (machine: assert 5). Therefore B5’s beyond-layer bound closes on the full admissible class WITHOUT any arbitrary- Q DR-2: the unrestricted-class bound $E_+(Q) \leq_\varepsilon N^{2+\varepsilon}$ (T-030) would only be needed to REMOVE the admissibility cap $T' \leq 10$ – i.e. to admit competitors that violate the θ_{min} coherence resolution. That is a FRONTIER STRENGTHENING of the competitor class, not a gap in B5’s admissibility-bounded statement.

Where the hypothesis IS load-bearing. H-NONLATTICE-REMAINDER-EXCLUDED is load-bearing ONLY for the strictly stronger statement in which H-ADM-COH (the θ_{min} admissibility itself) is DISCHARGED – removed as a hypothesis. That discharge (D1-A, 2026-06-12) used the lattice arithmetic (R-026/027/028) and holds for the lattice class; for non-lattice competitors the θ_{min} admissibility is RETAINED (it is the class definition, and Lemma 2 uses it). So the precise content of H-NONLATTICE-REMAINDER-EXCLUDED is: “the admissibility-DISCHARGED form of the bound is claimed for lattice competitors only.” It does NOT exclude non-lattice competitors from the admissibility-BOUNDED statement – those are covered by Lemma 2.

5. The operator-decision: reclassify the hypothesis

B5 has two statements of differing strength:

- (A) *Admissibility-bounded* (matches the published C_{full} head): the beyond-layer correction is below the layer margin for every admissible competitor. **Substantive hypotheses: none** – θ_{min} admissibility + packing are the DEFINITIONAL SCOPE, and Lemma 2 covers lattice + non-lattice.
- (B) *Admissibility-discharged* (strictly stronger, lattice-only): the bound holds even WITHOUT θ_{min} admissibility, via the lattice arithmetic. **H-NONLATTICE-REMAINDER-EXCLUDED is load-bearing here.**

The current B5 tier (T6 PROVED CONDITIONAL on {H-NONLATTICE-REMAINDER-EXCLUDED}) implicitly targets statement (B). If B5’s canonical statement is taken to be (A) – the one the published head uses – then the hypothesis set is EMPTY of substantive conditions (only definitional scope remains), which is an operator-decision tier question: (A) is a candidate T6/T7 with definitional scope and NO substantive conditional hypothesis. This note ENACTS nothing; it presents the reclassification for operator decision, exactly parallel to synthesis Sec. 4b for the head.

6. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) “Statement (A) just buries the hypothesis in the word ‘admissible’.” VALID and acknowledged: the θ_{min} admissibility is a real restriction on the competitor class. The point is that it is a DEFINITIONAL SCOPE (the physical coherence-resolution class, fixed by the microphysics), not a CONDITIONAL hypothesis pending a proof – the same status as ‘lattice shells’ in the head. Whether definitional scope warrants a tier above T6-conditional is the operator’s call; the note only removes the FALSE impression that arbitrary- Q DR-2 is a gap in (A).
- (β) “Lemma 2 needs $\rho \leq q_0$ – does every sum-circle satisfy it?” DISMISSED: $\rho = \sqrt{q_0^2 - |t|^2/4} \leq q_0$ for every t , with equality only at $t = 0$ (excluded); the bound is uniform over all proper sum-circles (machine assert 2).

(γ) “The realized max is 8, not the cap 10 – is the cap the right object?” DISMISSED: the bound only needs $T' \leq 10$ (which gives $R_{\text{lead}} \leq 0.510 < 1$); the realized 8 is even more comfortable. The cap is the rigorous upper bound; the realized value is a bonus margin.

Quantitative sanity check (mandatory). Lemma 2 cap = 10 at $\theta_{\min} = 0.596$; $R_{\text{lead}}(T'=10) = 23.2 \cdot 11 \cdot 2 \times 10^{-3} = 0.510 < 1$; $K(49) \approx 113 \ll 5972$; non-lattice worst occupancy = $2 \leq 10$ on random configs; all five checks PASS, exit 0. The arithmetic is the head’s own ($R_{\text{lead}} = 0.510$, $\times 1.96$ margin), reproduced here for the B5 machinery.

7. Result footer

Result ID:	B5-BEYOND-LAYER-BOUND / route-3 non-load-bearing analysis
Precise statement:	B5’s beyond-layer bound depends on the competitor only through (T’, n). The coherence circle-packing Lemma 2 (pure geometry, no lattice/DR-2/decoupling) caps $T' \leq \text{floor}(2\pi/\theta_{\min}) \leq 10$ for EVERY admissible competitor, lattice OR non-lattice, and $n \leq n_{\text{pack}}$. Hence the admissibility-bounded statement (A) closes on the full admissible class with NO arbitrary-Q DR-2; H-NONLATTICE-REMAINDER-EXCLUDED is NON-LOAD-BEARING for (A), load-bearing ONLY for the stronger admissibility-discharged lattice-only (B).
Scope:	B5 hypothesis analysis; structural, no new theorem (Lemma 2 = res5_036), no flip.
Evidence grade:	STRUCTURAL ANALYSIS + EXECUTED (dr2_t030_route3_nonloadbearing.py 5/5).
Reproduction command:	python codes/vacuum/dr2_t030_route3_nonloadbearing.py
Falsifier:	an admissible competitor with sum-circle occupancy $> \text{floor}(2\pi/\theta_{\min})$; or a beyond-layer quantity depending on the competitor beyond (T’, n).
Tier before / after:	none -- operator decides. Candidate reclassification: H-NONLATTICE-REMAINDER-EXCLUDED from CONDITIONAL hypothesis to DEFINITIONAL SCOPE for statement (A); B5 stays T6-conditional until the operator rules on (A) vs (B).
No-overclaim statement:	arbitrary-Q DR-2 (T-030) stays OPEN as a frontier strengthening (removing the admissibility cap); this note does NOT close it and does NOT flip B5; statement (A)’s admissibility is definitional scope, not a discharged hypothesis.
Next required action:	operator: rule whether B5’s canonical statement is (A) admissibility-bounded (hypothesis set empty of substantive conditions -> candidate tier above T6-conditional) or (B) admissibility-discharged (H-NONLATTICE load-bearing, lattice-only); reconcile with the synthesis Sec. 4b head finding.